

1 Solution — GIAM 1.1

Problem 4

1.1 Problem

The “see and say” sequence is produced by first writing 1, then iterating the following procedure: *look at the previous entry and say how many entries there are of each integer; write down what you just said*. The first several terms are

$$1, 11, 21, 1112, 3112, 211213, 312213, 212223, \dots$$

Comment on the rationality of the number

$$N = 0.111211123112211213\dots$$

obtained by concatenating the decimal digits of the sequence.

1.2 Verifying the Rule

The rule is: read the previous term, count how often each digit 1, 2, 3, ... appears in it (ignoring digits that don’t appear), and write the result as alternating (*count*, *digit*) pairs in ascending order of digit.

Step	Term	Count of 1	of 2	of 3	of 4	New term
1	1	1	—	—	—	11
2	11	2	—	—	—	21
3	21	1	1	—	—	1112
4	1112	3	1	—	—	3112
5	3112	2	1	1	—	211213
6	211213	3	2	1	—	312213
7	312213	2	2	2	—	212223

Table 1.1

So the first eight terms match the problem’s list. Now extend.

1.3 Extending the Sequence

Step	Term	Count of 1	of 2	of 3	of 4	New term
8	212223	1	4	1	—	114213
9	114213	3	1	1	1	31121314
10	31121314	4	1	2	1	41122314
11	41122314	3	2	1	2	31221324
12	31221324	2	3	2	1	21322314
13	21322314	2	3	2	1	21322314

Table 1.2

The sequence **reaches a fixed point at step 12** — every subsequent term is also **21322314**.

1.4 The Fixed Point — Why 21322314 Describes Itself

Read **21322314** as four “count-digit” pairs: **21**, **32**, **23**, **14**. That says: “two 1s, three 2s, two 3s, one 4.” Counting the digits in **21322314** itself confirms each claim:

"21322314" describes itself: each pair "count digit" reports a true count in the string

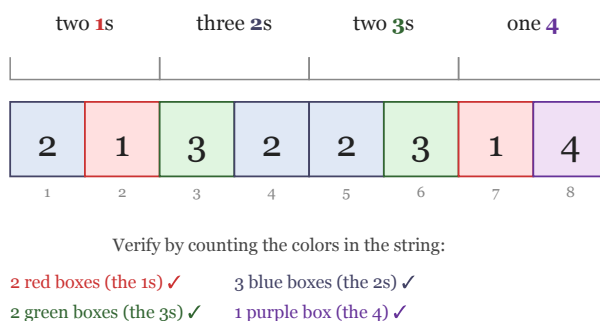


Figure 1.1: Each of the four count-digit pairs labels a true count of the matching digit in the string. The string describes itself — so applying the see-and-say rule returns the same string.

digit	count in 21322314	claim of the matching pair
1	2	“two 1s” (the pair 21)
2	3	“three 2s” (the pair 32)
3	2	“two 3s” (the pair 23)
4	1	“one 4” (the pair 14)

Table 1.3

All four claims hold, so **21322314** is **its own description**. The see-and-say map fixes it. ■

1.5 Concatenated Decimal N — Eventually Periodic

Let L be the total length, in digits, of the first 12 terms (everything before the sequence stabilises). Summing the lengths:

$$L = \underbrace{1}_{t_1} + \underbrace{2}_{t_2} + \underbrace{2}_{t_3} + \underbrace{4}_{t_4} + \underbrace{4}_{t_5} + \underbrace{6}_{t_6} + \underbrace{6}_{t_7} + \underbrace{6}_{t_8} + \underbrace{6}_{t_9} + \underbrace{8}_{t_{10}} + \underbrace{8}_{t_{11}} + \underbrace{8}_{t_{12}} =$$

From decimal position $L + 1 = 62$ onward, N consists of the 8-digit block **21322314** repeated forever:

$$N = 0. \underbrace{111211123112211213312213212223114213311213141112231431221324}_{\text{first 61 digits — the pre-period}} \overline{21322314}.$$

The decimal is *eventually periodic* — pre-period of length 61, period of length 8 — so

$$N \in \mathbb{Q}.$$

The number is **rational**. ■

1.6 Remark — Comparison to Conway’s Standard Look-and-Say

The textbook’s variant is *not* John Conway’s classical look-and-say sequence. Conway’s version describes *consecutive runs*: $1 \rightarrow 11 \rightarrow 21 \rightarrow 1211 \rightarrow 111221 \rightarrow 312211 \rightarrow \dots$, where (for instance) **1211** reads “one 1, one 2, two 1s” because the two consecutive **1s** at the end form a single run.

Conway proved a famous result about his sequence: the lengths grow asymptotically like λ^n where $\lambda \approx 1.30357\dots$ is “Conway’s constant” — the unique real root of a degree-71 polynomial. The standard look-and-say sequence *never* stabilises, and its concatenation is irrational (in fact transcendental).

Our textbook variant is different — it counts each *distinct digit* (not consecutive runs) and orders the pairs by digit value. The resulting dynamics are completely different: instead of exponential growth, the sequence rapidly *collapses to a fixed point*, and the concatenated decimal is rational.

1.7 Remark — Why a Fixed Point Forces Rationality

Once the sequence stabilises at any string s of length ℓ , the concatenated decimal from that point onward is

$$s s s s s \dots = \overline{s}$$

— literally the repeating block s with period ℓ . The earlier (non-stabilised) terms form a finite pre-period. *Any* eventually-stabilising sequence will produce a rational concatenation, regardless of how long the pre-period is.

So the question “is N rational?” reduces to “does the see-and-say sequence (in this variant, starting from 1) reach a fixed point?” The trace above answers: yes, at step 12.

1.8 Remark — Other Fixed Points

21322314 isn’t the only fixed point of this see-and-say map. The simplest other one is

$$22 = \text{"two 2s"},$$

a 2-digit self-describing string. Any string that already describes its own digit counts will be fixed.

Other examples (using larger digit sets):

- **213222314** (with a "0-count" slot) — *not* a valid fixed point in our convention because we skip absent digits.
- **21322314** $\underbrace{15}_{\text{counting 5s}}$ would describe “two 1s, three 2s, two 3s, one 4, one 5” — i.e. a 10-digit string **2132231415**. Is it a fixed point? Let me check: digits of **2132231415** — 1 appears 3 times (positions 2, 7, 9), 2 appears 3 times (positions 1, 4, 5), 3 appears 2 times (positions 3, 6), 4 appears 1 time (position 8), 5 appears 1 time (position 10). So the new term would be “3, 1, 3, 2, 2, 3, 1, 4, 1, 5” = “3132231415” — *not* the same. So **2132231415** is *not* a fixed point.

Finding all fixed points is an interesting combinatorial puzzle in its own right. For sequences starting from **1**, the dynamics happen to land at **21322314** — but a different starting string can fix at **22** or could in principle land in a longer cycle.

1.9 Remark — A Tighter View of N

Since N has pre-period 61 and period 8, we can write it as a rational number. Let A be the integer formed by the first 61 digits and $B = \mathbf{21322314}$ the period block:

$$N = \frac{A}{10^{61}} + \frac{1}{10^{61}} \cdot \frac{B}{10^8 - 1} = \frac{A(10^8 - 1) + B}{10^{61}(10^8 - 1)}.$$

The denominator is

$$10^{61}(10^8 - 1) = 10^{61} \cdot 99999999.$$

We won't simplify A explicitly (it's a 61-digit integer), but this *is* the canonical “fraction in lowest pre-period + period form”. The denominator's relevant factor is $99999999 = 10^8 - 1$ — the constant that always shows up for a period-8 repeating decimal.

1.10 Remark — How the Problem Would Read with Conway's Standard Variant

If the textbook had used Conway's classical look-and-say sequence — describing consecutive *runs* of digits — the answer would be **irrational** (and famously, transcendental). The sequence

1, 11, 21, 1211, 111221, 312211, 13112221, 1113213211, ...

grows without bound (lengths blow up by $\lambda \approx 1.30357\dots$ each step), never repeats, and the concatenation $0.1112112111221312211\dots$ has no eventually-periodic structure.

The textbook's choice of the count-each-distinct-digit variant is what makes the answer rational. It's a nice illustration of how sensitive the rationality of “weird” decimals is to the precise rule generating them.